Applying componendo and dividendo rule, we get

$$\frac{\sin A\cos B+\cos A\sin B}{\sin A\cos B-\cos A\sin B}=\frac{m+1}{m-1}$$

$$\frac{\sin(A+B)}{\sin(A-B)}=\frac{m+1}{m-1},$$

which completes the proof.

### 4.2.3 Trigonometric ratios of multiple angles

Identities involving sin2$A$, cos2$A$, tan2$A$, sin3$A$ etc., are called multiple angle identities.

(i) $\sin 2A=\sin(A+A)=\sin A\cos A+\cos A\sin A=2\sin A\cos A.$

(ii) $\cos 2A=\cos(A+A)=\cos A\cos A-\sin A\sin A=\cos^2 A-\sin^2 A$

(iii) $\sin 3A=\sin(2A+A)$

$=\sin 2A\cos A+\cos 2A\sin A$

$=(2\sin A\cos A)\cos A+(1-2\sin^2 A)\sin A$

$=2\sin A\cos^2 A+\sin A-2\sin^3 A$

$=2\sin A(1-\sin^2 A)+\sin A-2\sin^3 A$

$=3\sin A-4\sin^3 A$

Thus we have the following multiple angles formulae

1. (i) $\sin 2A=2\sin A\cos A$

   (ii) $\sin 2A=\dfrac{2\tan A}{1+\tan^2 A}$

2. (i) $\cos 2A=\cos^2 A-\sin^2 A$

   (ii) $\cos 2A=2\cos^2 A-1$

   (iii) $\cos 2A=1-2\sin^2 A$

   (iv) $\cos 2A=\dfrac{1-\tan^2 A}{1+\tan^2 A}$

3. $\tan 2A=\dfrac{2\tan A}{1-\tan^2 A}$

4. $\sin 3A=3\sin A-4\sin^3 A$

5. $\cos 3A=4\cos^3 A-3\cos A$

6. $\tan 3A=\dfrac{3\tan A-\tan^3 A}{1-3\tan^2 A}$

**NOTE**

(i) $\cos^2 A=\frac{1}{2}(1+\cos 2A)$ (or)

$\cos A=\pm\sqrt{\dfrac{1+\cos 2A}{2}}$

(ii) $\sin^2 A=\frac{1}{2}(1-\cos 2A)$ (or)

$\sin A=\pm\sqrt{\dfrac{1-\cos 2A}{2}}$

**Example 4.9**

Show that $\dfrac{\sin 2\theta}{1+\cos 2\theta}=\tan\theta$

***Solution***

$$\frac{\sin 2\theta}{1+\cos 2\theta}=\frac{2\sin\theta\cos\theta}{2\cos^2\theta}=\frac{\sin\theta}{\cos\theta}=\tan\theta$$

**Example 4.10**

Using multiple angle identity, find tan60°

***Solution***

$$\tan 2A=\frac{2\tan A}{1-\tan^2 A}$$

Put A = 30° in the above identity, we get

$$\tan 60^\circ=\frac{2\tan 30^\circ}{1-\tan^2 30^\circ}$$

$$=\frac{2\frac{1}{\sqrt{3}}}{1-\frac{1}{3}}=\frac{\frac{2}{\sqrt{3}}}{\frac{2}{3}}=\frac{2}{\sqrt{3}}\times\frac{3}{2}=\sqrt{3}$$

**Example 4.11**

If $\tan A=\frac{1}{7}$ and $\tan B=\frac{1}{3}$, show that cos2$A$ = sin4$B$

***Solution***

$$\cos 2A=\frac{1-\tan^2 A}{1+\tan^2 A}=\frac{1-\frac{1}{49}}{1+\frac{1}{49}}=\frac{48}{49}\times\frac{49}{50}$$

$$=\frac{24}{25} \qquad \ldots (1)$$



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